latexkit — Gallery

Thành phần dùng chung cho ghi chú, paper, báo cáo, sách và slide

10 tháng 11, 2026

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1 Cài đặt trong tài liệu

```
\usepackage[lang=vi, table, diagram, plot, code, box, math, algo, science]{latexkit}
\usepackage[lang=en, all]{latexkit}          % every module, English labels
\usepackage[style=plain, accent=8B0000]{latexkit}  % classic headings, red accent
```

2 Bảng (table)

```
\begin{lktable}[H]{Kết quả chính}{tab:demo}{l ccc}
  Phương pháp & PSNR\up & LPIPS\down & Real time \\ \midrule
  NeRF        & 26.1          & 0.31          & \xmark \\
  3DGS        & \second{27.4} & \second{0.22} & \cmark \\
  \lkhl Ours  & \best{28.8}   & \best{0.18}   & \cmark \\
\end{lktable}
```

Bảng 1: Kết quả chính

Phương pháp	PSNR↑	LPIPS↓	Real time
NeRF	26.1	0.31	✗
3DGS	27.4	0.22	✓
Ours	**28.8**	**0.18**	✓

```
\begin{lktablex}[H]{Bảng giãn hết chiều ngang, ô màu theo giá trị}{tab:heat}{l C C C}
  Tiêu chí & A & B & C \\ \midrule
  Chất lượng & \lkheat{0.6} & \lkheat{0.9} & \lkheat{0.4} \\
  Tốc độ     & \lkbar{0.8}  & \lkbar{0.5}  & \lkbar{0.9} \\
  LiDAR      & \cmark & \pmark & \xmark \\
\end{lktablex}
```

Bảng 2: Bảng giãn hết chiều ngang, ô màu theo giá trị

Tiêu chí	A	B	C
Chất lượng	0.6	0.9	0.4
Tốc độ			
LiDAR	✓	∼	✗

```
\begin{center}
\begin{lktblr}{colspec={l S[table-format=2.1] S[table-format=1.3]}}
  {Method} & {PSNR} & {SSIM} \\
  Baseline & 25.1 & 0.810 \\
  Ours     & 28.8 & 0.893 \\
\end{lktblr}
\end{center}
```

Bảng 3:

Method	PSNR	SSIM
Baseline	25,1	0,810
Ours	28,8	0,893

2.1 Đồ họa dữ liệu trong bảng (gradbars)

```
\begin{gradtable}[width=\linewidth, header align=c]
  \gradcolumn{method}{Phương pháp}
  \gradcolumn[type=bar, mark=both, options={max=100, precision=1}]{score}{Điểm}
  \gradcolumn[type=number, mark=best, options={better=lower, precision=1}]{time}{Thời gian / ms}
  \gradrow{NeRF, 72.4, 18.2}
  \gradrow{3DGS, 92.7, 11.5}
  \gradrow{Ours, 88.1, 9.6}
\end{gradtable}

Trong câu: \gradbar[max=100, width=25mm]{72.4} \quad \gradspark{3,5,2,8,6}
\quad \gradheat[min=0, max=1]{0.7} \quad \gradstack{30,50,20}
```

Phương pháp	Điểm	Thời gian / ms
NeRF	72.4	18.2
3DGS	**92.7**	11.5
Ours	88.1	**9.6**

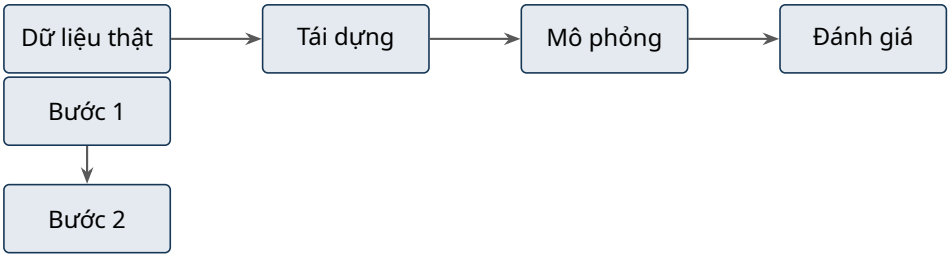
Trong câu: 72.4 6.0 0.7 100.0

3 Sơ đồ (diagram)

3.1 Pipeline và timeline nhanh

```
\lkpipeline{Dữ liệu thật, Tái dựng, Mô phỏng, Đánh giá}

\lkpipeline*[node distance=5mm]{Bước 1, Bước 2}
```

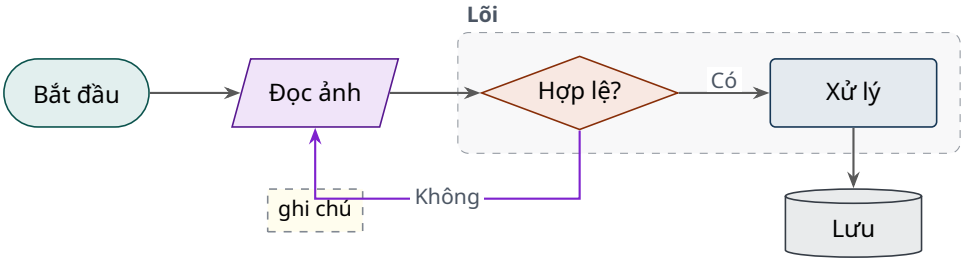



```
\lktimeline{2020/NeRF, 2022/Instant-NGP, 2023/3DGS, 2024/{Street Gaussians}}
```

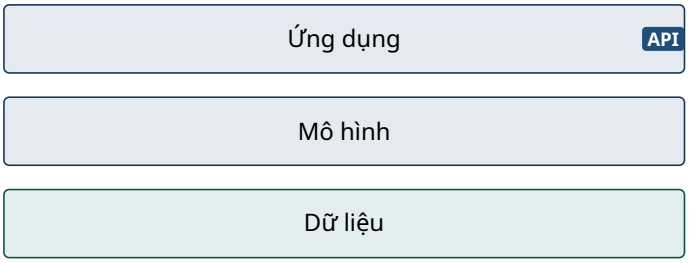


3.2 Các style node và cạnh

```
\begin{tikzpicture}[lk]
  \node[terminal]                (s) {Bắt đầu};
  \node[io,        right=of s]   (i) {Đọc ảnh};
  \node[decision,  right=of i]   (d) {Hợp lệ?};
  \node[process,   right=of d]   (p) {Xử lý};
  \node[datastore, below=of p]   (db) {Lưu};
  \node[note,      below=of i]   (n) {ghi chú};
  \draw[flow] (s) -- (i);  \draw[flow] (i) -- (d);
  \draw[flow] (d) -- node[elabel, above] {Có} (p);
  \draw[flow] (p) -- (db);
  \draw[loop] (d.south) |- ++(0,-0.9) -| node[elabel, pos=0.25] {Không} (i.south);
  \lkgroup{(d)(p)}{Lõi}
\end{tikzpicture}
```

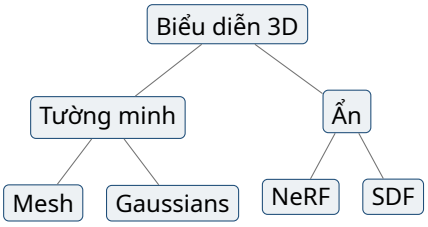



```
\begin{tikzpicture}[lk]
  \node[layer=9cm]                                (a) {Ứng dụng};
  \node[layer=9cm, below=3mm of a]                (b) {Mô hình};
  \node[layer=9cm, below=3mm of b, lkcolor=lkteal] (c) {Dữ liệu};
  \node[tag, anchor=east] at (a.east) {API};
\end{tikzpicture}
```



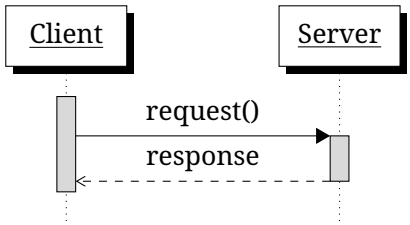
3.3 Cây phân loại (forest)

```
\begin{forest}
  for tree={draw=accent, fill=accent!8, rounded corners=2pt, font=\sffamily\small,
            edge={draw=black!55}, l sep=6mm, s sep=3mm}
  [Biểu diễn 3D [Tường minh [Mesh] [Gaussians]] [Ẩn [NeRF] [SDF]]]
\end{forest}
```



3.4 Sơ đồ tuần tự (pgf-umlsd)

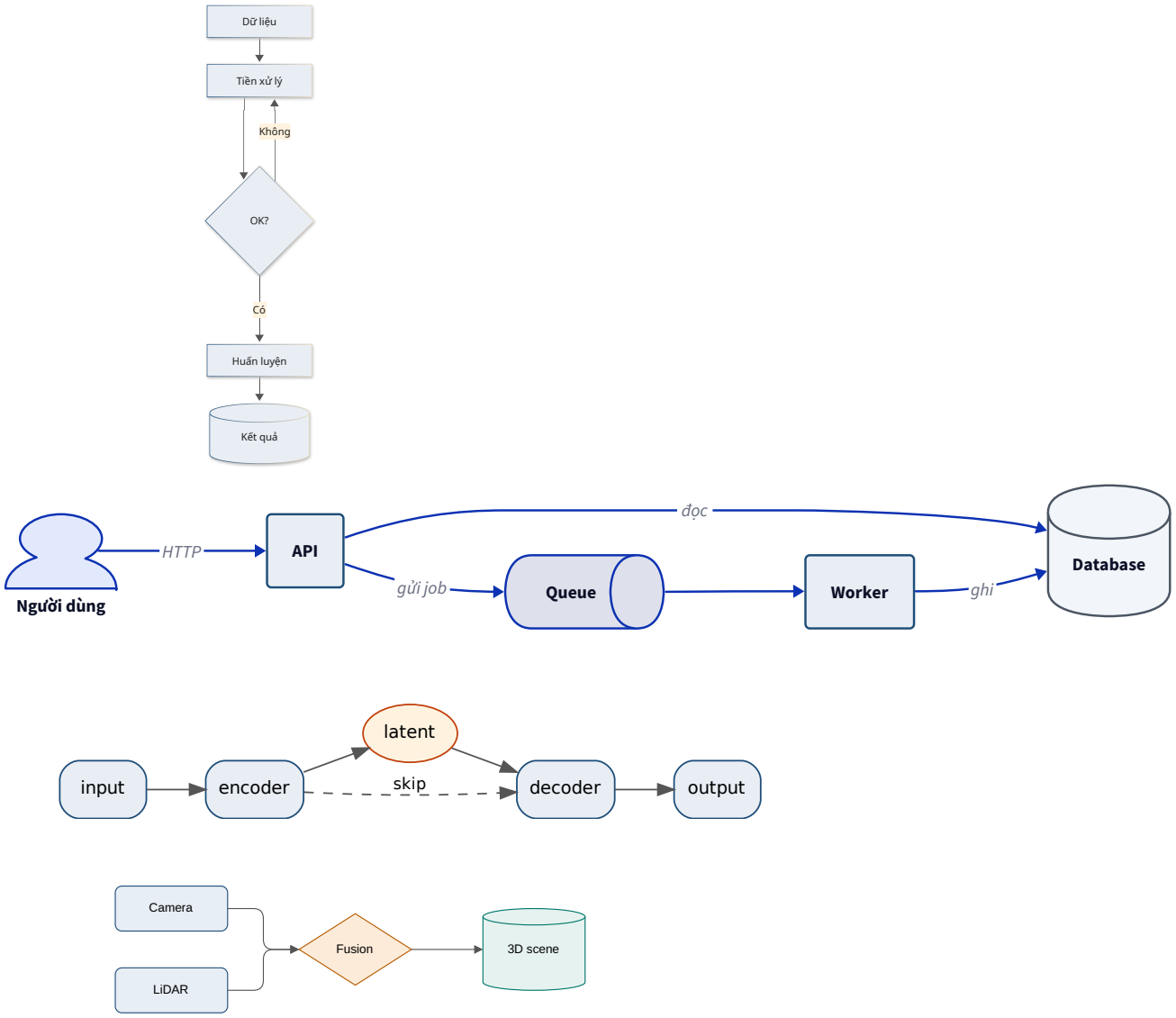
```
\begin{sequencediagram}
  \newthread{c}{Client}
  \newinst[2]{s}{Server}
  \begin{call}{c}{request()}{s}{response}\end{call}
\end{sequencediagram}
```



3.5 Sơ đồ từ file: Mermaid, D2, Graphviz, draw.io

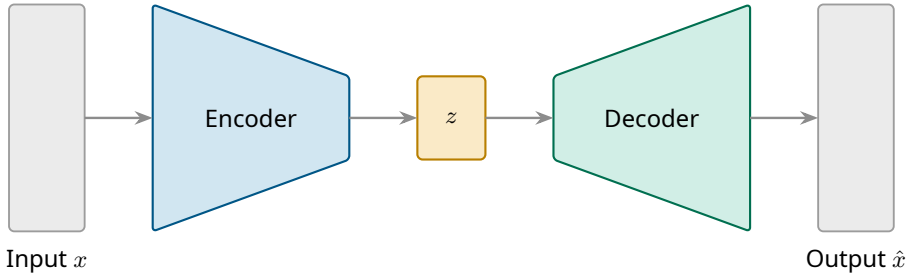
File nằm trong `diagrams/`; latexmk tự chuyển sang PDF.

```
\lkdiagram[max height=7cm]{diagrams/pipeline}   % pipeline.mmd
\lkdiagram{diagrams/architecture}                % architecture.d2
\lkdiagram[max width=0.7\linewidth]{diagrams/graph}  % graph.dot
\lkdiagram[max width=0.6\linewidth]{diagrams/sketch} % sketch.drawio
```



3.6 Hình OpenTikZ

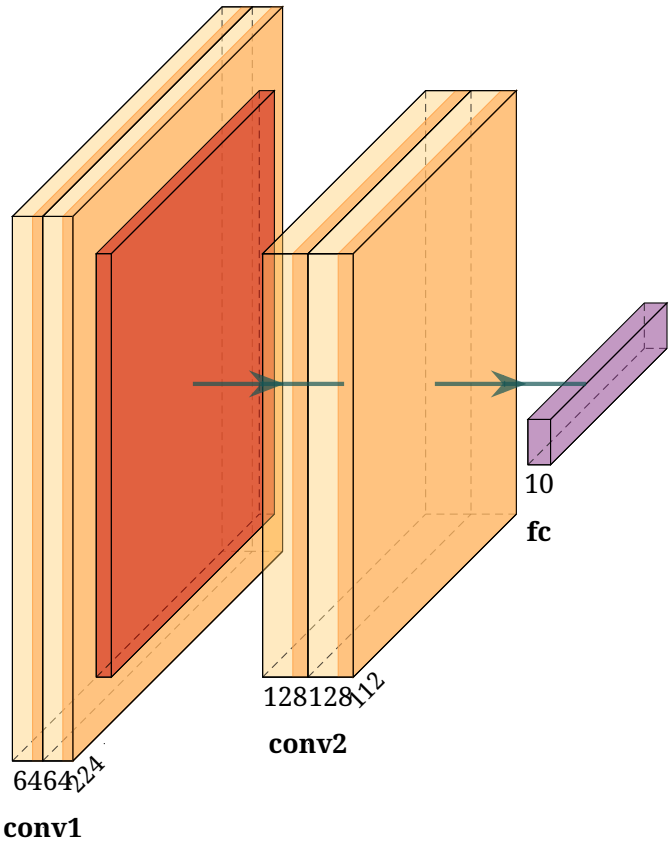
```
\lktikzfig[max height=6cm]{../assets/vendor/opentikz/templates/encoder-decoder/template}
```



Thêm vào project: `latexkit asset add encoder-decoder` rồi `\lktikzfig{figures/encoder-decoder}`. Toàn bộ thư viện ở Mục 8.

3.7 Kiến trúc mạng nơ-ron (PlotNeuralNet)

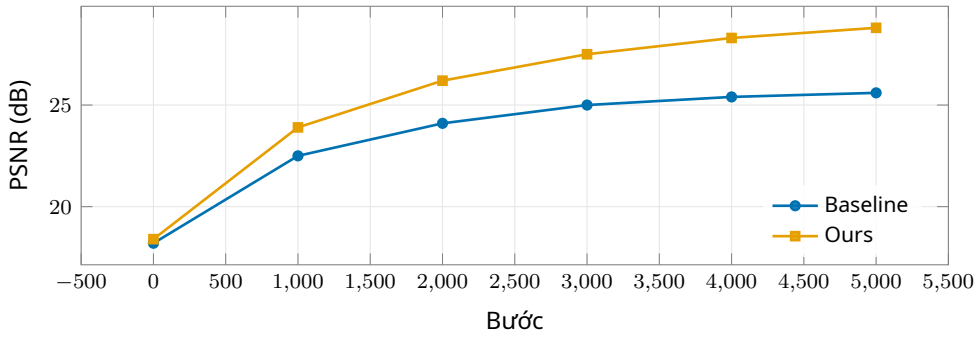
```
\begin{tikzpicture}
  \pic[shift={(0,0,0)}] at (0,0,0) {RightBandedBox={name=c1, caption=conv1,
       xlabel={{"64","64"}}, zlabel=224, fill=\ConvColor, bandfill=\ConvReluColor,
       height=36, width={2,2}, depth=36}};
  \pic[shift={(0,0,0)}] at (c1-east) {Box={name=p1, fill=\PoolColor, opacity=0.5,
       height=28, width=1, depth=28}};
  \pic[shift={(2,0,0)}] at (p1-east) {RightBandedBox={name=c2, caption=conv2,
       xlabel={{"128","128"}}, zlabel=112, fill=\ConvColor, bandfill=\ConvReluColor,
       height=28, width={3,3}, depth=28}};
  \pic[shift={(2,0,0)}] at (c2-east) {Box={name=fc, caption=fc, xlabel={{"10",}},
       fill=\SoftmaxColor, height=3, width=1.5, depth=20}};
  \draw[connection] (p1-east) -- node{\midarrow} (c2-west);
  \draw[connection] (c2-east) -- node{\midarrow} (fc-west);
\end{tikzpicture}
```



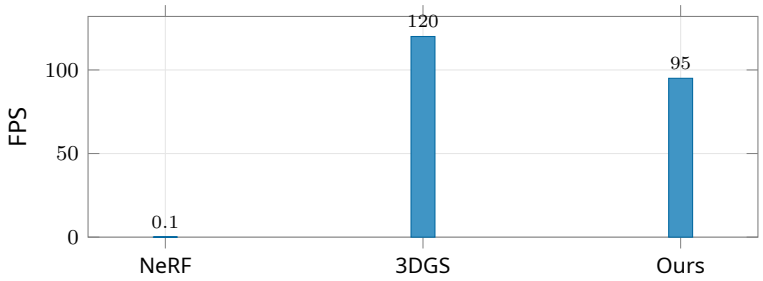
Hình có sẵn: `latexkit asset list nn` (AlexNet, VGG16, U-Net, FCN, HED...).

4 Đồ thị (plot)

```
\begin{tikzpicture}
  \begin{axis}[lkplot, width=0.75\linewidth, height=5cm, xlabel={Bước}, ylabel={PSNR (dB)},
               legend pos=south east]
    \addplot table[x=step, y=baseline, col sep=comma] {../templates/_common/plots/results.csv};
    \addplot table[x=step, y=ours, col sep=comma]     {../templates/_common/plots/results.csv};
    \legend{Baseline, Ours}
  \end{axis}
\end{tikzpicture}
```




```
\begin{tikzpicture}
  \begin{axis}[lkbar, symbolic x coords={NeRF, 3DGS, Ours}, ylabel={FPS}, ymin=0,
               width=0.6\linewidth, height=4.5cm]
    \addplot coordinates {(NeRF, 0.1) (3DGS, 120) (Ours, 95)};
  \end{axis}
\end{tikzpicture}
```



Đồ thị bằng Python dùng style SciencePlots có sẵn trong `plots/styles`:

Đoạn code 1: plots/make_plots.py

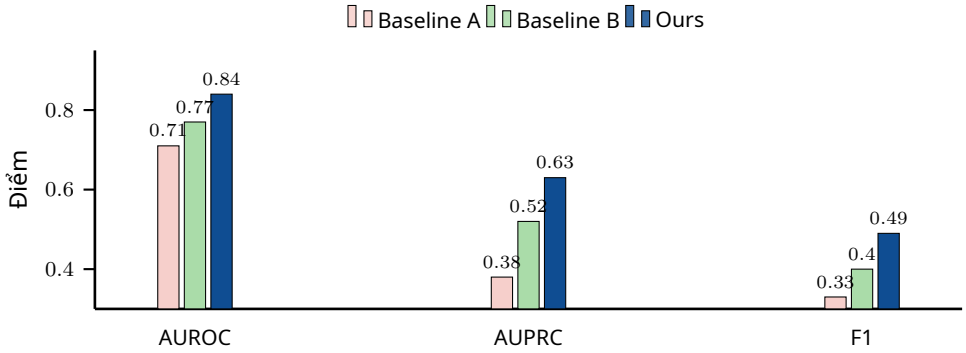
```python
from latexkit_plot import use_style, save
use_style("science", "ieee")      # hoặc "nature", "grid", "no-latex", ...
fig, ax = plt.subplots()
ax.plot(step, psnr)
save(fig, "psnr")                 # -> figures/psnr.pdf, chèn bằng \lkfig{psnr}{...}{...}
```

4.1 Phong cách figures4papers

Bảng màu và kiểu trình bày lấy từ figures4papers (Chen Liu, CC BY-NC 4.0): xanh đậm cho phương pháp đề xuất, xanh lá cho cải tiến, đỏ hồng cho baseline, chỉ giữ trục trái và trục dưới. Dùng được cho cả pgfplots (`lkf4p`, `lkf4pbar`, màu `f4pblue`, `f4pgreenA..C`, `f4predA..B`, `f4pred`, `f4pgray`) và matplotlib (`plots/f4p.py`).

```
\begin{tikzpicture}
  \begin{axis}[lkf4pbar, symbolic x coords={AUROC, AUPRC, F1}, ylabel={Điểm}, ymin=0.3, ymax=0.95,
               width=0.75\linewidth, height=5cm, bar width=8pt, legend columns=3,
```

```
            legend style={at={(0.5,1.03)}, anchor=south}]
    \addplot coordinates {(AUROC, 0.71) (AUPRC, 0.38) (F1, 0.33)};
    \addplot coordinates {(AUROC, 0.77) (AUPRC, 0.52) (F1, 0.40)};
    \addplot[fill=f4pblue, draw=black] coordinates {(AUROC, 0.84) (AUPRC, 0.63) (F1, 0.49)};
    \legend{Baseline A, Baseline B, Ours}
  \end{axis}
\end{tikzpicture}
```



Bằng Python: `latexkit asset add f4p-bars-comparison` chép script mẫu vào `plots/`, sửa số liệu ở đầu file rồi chạy `latexkit plot`. Có 12 mẫu (`latexkit asset list plot`), xem ở Mục 8.

Đoạn code 2: plots/ket_qua.py

```python
from f4p import PALETTE, LARGE, apply_publication_style, make_grouped_bar, finalize_figure
apply_publication_style(LARGE)            # font 24, trục dày 3: hình so sánh lớn
fig, ax = plt.subplots(figsize=(12, 5))
make_grouped_bar(ax, ["A", "B"], [[0.78, 0.80], [0.92, 0.88]], ["Baseline", "Ours"],
                 colors=[PALETTE["red_1"], PALETTE["blue_main"]], hatches=["//", ""], annotate=
    True)
finalize_figure(fig, "ket_qua")           # -> figures/ket_qua.pdf
```

5 Code (code)

Đoạn code 3: Vòng lặp huấn luyện

```python
for epoch in range(num_epochs):     # chú thích tiếng Việt vẫn hiển thị đúng
    loss = criterion(model(x), y)
    loss.backward()
```

Viết là `\begin{lkcode}[Python][label=lst:train]{Tiêu đề}`; tham chiếu bằng Đoạn code 3. Lấy code từ file: `\lkcodefile[Python]{code/train.py}`. Lệnh terminal:

```
$ latexkit new report bao-cao -t "Báo cáo" -m all
```

6 Hộp ghi chú (box)

```
\begin{keyidea} Ý chính của phần. \end{keyidea}
\begin{info} Thông tin thêm. \end{info}
\begin{tip} Mẹo. \end{tip}
\begin{warning} Lưu ý. \end{warning}
\begin{question}[Câu hỏi khi đọc] Paper giải quyết vấn đề gì? \end{question}
\begin{examplebox} Ví dụ. \end{examplebox}
\begin{summary} Tóm tắt. \end{summary}
```

Ý chính

Ý chính của phần.

Ghi chú

Thông tin thêm.

Mẹo

Mẹo.

Lưu ý

Lưu ý.

Câu hỏi khi đọc

Paper giải quyết vấn đề gì?

Ví dụ

Ví dụ.

Tóm tắt

Tóm tắt.

7 Toán và thuật toán (`math`, `algo`, `science`)

```
\begin{definition} Hàm $f:\R^n\to\R$ lồi nếu $f(\lambda x+(1-\lambda)y)\le\lambda f(x)+(1-\lambda)f(y)$. \end{
    definition}
$\hat\theta = \argmin_\theta \norm{\mat{A}\vect{x}-\vect{b}}^2$, \quad \SI{9.81}{\metre\per\second\squared}, \
    quad \num{12345.678}
```

Định nghĩa 7.1. Hàm $f:\mathbb{R}^n\to\mathbb{R}$ lồi nếu $f(\lambda x+(1-\lambda)y)\le\lambda f(x)+(1-\lambda)f(y)$.
$\hat\theta = \arg\min_\theta \|\mathbf{A}\boldsymbol{x}-\boldsymbol{b}\|^2$, $\quad 9{,}81\,\mathrm{m\,s^{-2}}$, $\quad 12\,345{,}678$

```
\begin{algorithm}[H]
  \caption{Gradient descent}
  \begin{algorithmic}[1]
    \Require $\theta_0, \eta$
    \While{chưa hội tụ}
      \State $\theta \gets \theta - \eta\nabla\mathcal{L}(\theta)$ \Comment{cập nhật}
    \EndWhile
  \end{algorithmic}
\end{algorithm}
```

Thuật toán 1 Gradient descent

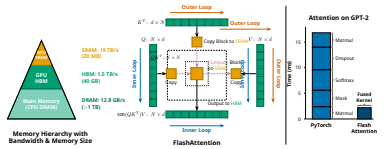
Đầu vào: θ_0, η

1: **while** chưa hội tụ **do**
2: $\quad \theta \leftarrow \theta - \eta\nabla\mathcal{L}(\theta)$ ▷ cập nhật
3: **end while**

8 Thư viện hình: OpenTikZ, PlotNeuralNet và figures4papers

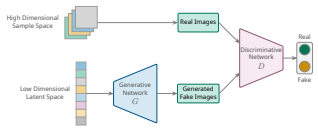
Mỗi ô là một file `standalone` trong `assets/vendor/opentikz` (hình theo CC0 1.0, code theo MIT) hoặc `assets/vendor/plotneuralnet` (MIT, loại `nn`), biên dịch cùng latexkit, hoặc một script matplotlib trong `assets/vendor/figures4papers` (CC BY-NC 4.0: ghi nguồn, không dùng cho mục đích thương mại; loại `plot`, số liệu chỉ là ví dụ). Thêm vào project bằng `latexkit asset add <id>`.

flash-attention example



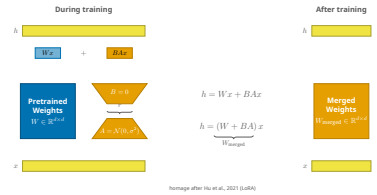
FlashAttention (IO-aware attention)

gan example



Generative Adversarial Network (GAN)

lora example



LoRA (low-rank adaptation)

anthropic icon

Anthropic

arxiv icon



arXiv

attention icon



Attention block

claude icon



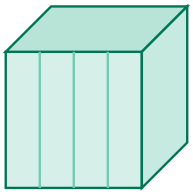
Claude

cloud icon



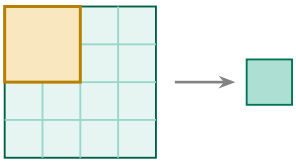
Cloud

container icon



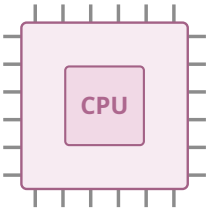
Container

convolution icon



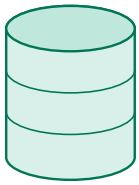
Convolution

cpu icon



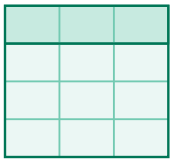
CPU

database icon



Database

dataset icon



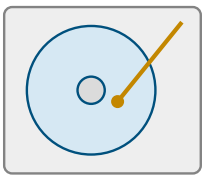
Dataset / table

deepseek icon



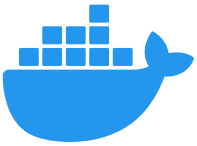
DeepSeek

disk icon



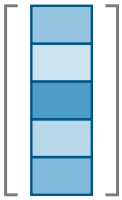
Disk / storage

docker icon



Docker

embedding icon



Embedding vector

git icon



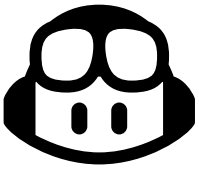
Git

github icon



GitHub

github-copilot icon



GitHub Copilot

google icon



Google

google-cloud icon



Google Cloud

google-colab icon



Google Colab

google-gemini icon



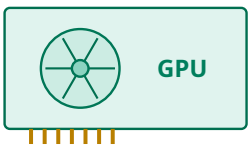
Google Gemini

google-scholar icon



Google Scholar

gpu icon



GPU

huggingface icon



Hugging Face

ieee icon



IEEE

jupyter icon



Jupyter

kaggle icon



Kaggle

keras icon



Keras

kubernetes icon



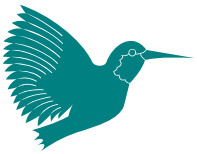
Kubernetes

langchain icon



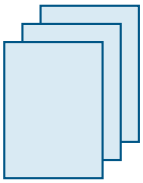
LangChain

latex icon



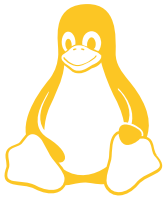
LaTeX

layer icon



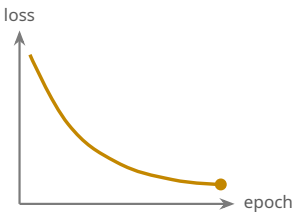
Layer / feature maps

linux icon



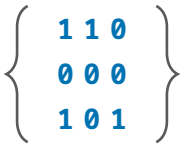
Linux

loss icon



Loss curve

matrix icon



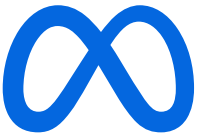
Binary matrix

mcp icon



Model Context Protocol

meta icon



Meta

mistral icon



Mistral AI

mobile icon



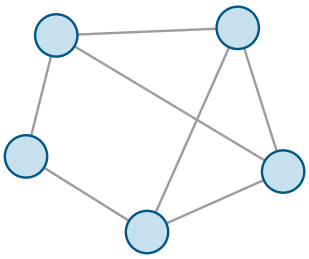
Mobile / edge device

model icon



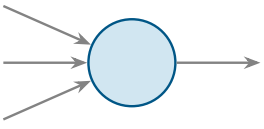
Model

network icon



Network

neuron icon



Neuron

numpy icon



NumPy

nvidia icon



NVIDIA

ollama icon



Ollama

orcid icon



ORCID

overleaf icon



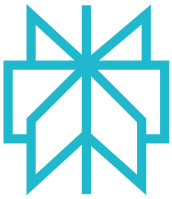
Overleaf

pandas icon



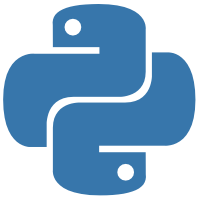
pandas

perplexity icon



Perplexity

python icon



Python

pytorch icon



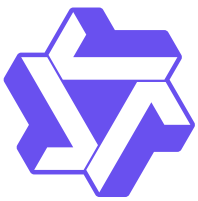
PyTorch

queue icon



Message queue

qwen icon



Qwen

reddit icon



Reddit

scikit-learn icon



scikit-learn

semantic-scholar icon



Semantic Scholar

server icon



Server

tensorflow icon



TensorFlow

user icon



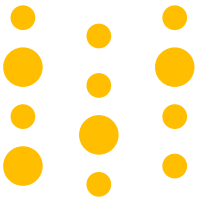
User

vllm icon



vLLM

wandb icon



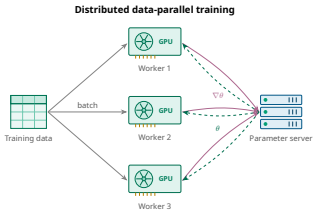
Weights & Biases

agent-mcp-stack template



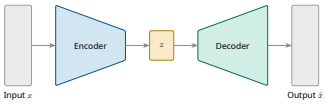
Agent Tool-Use Stack (MCP)

distributed-training template



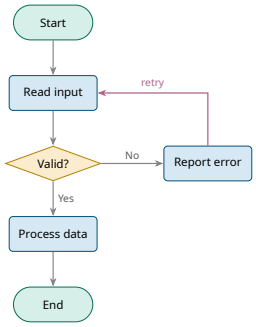
Distributed data-parallel training

encoder-decoder template



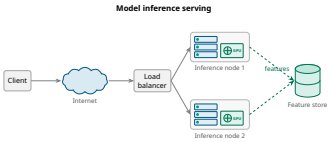
Encoder-decoder (bottleneck)

flowchart template



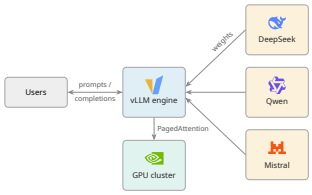
Flowchart (decision + loop)

inference-serving template



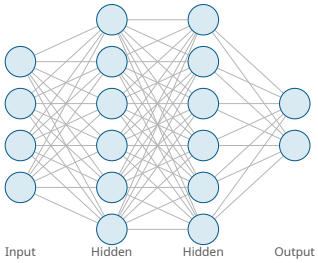
Model inference serving

llm-serving-stack template



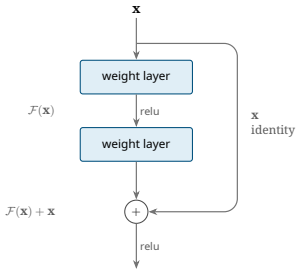
LLM Serving Stack

neural-net template



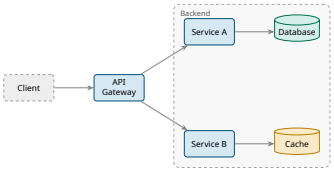
Feed-forward neural network

resnet-block template



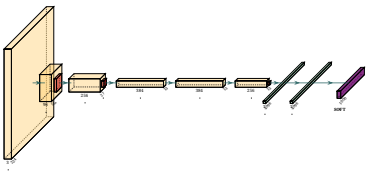
ResNet residual block

system-block-diagram template



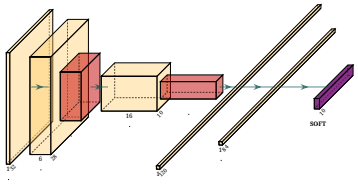
System block diagram

nn-alexnet nn



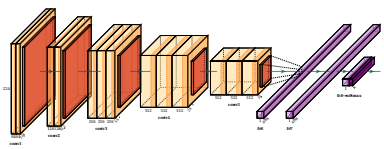
AlexNet

nn-lenet nn



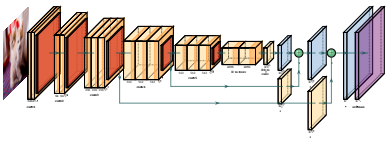
LeNet

nn-vgg16 nn



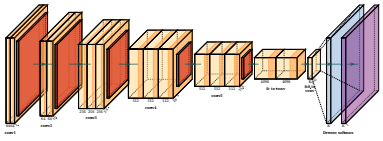
VGG16

nn-fcn8 nn



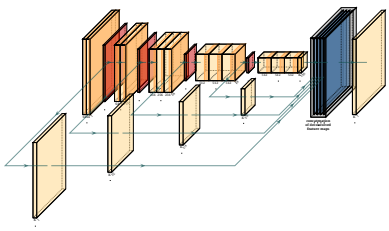
FCN-8s

nn-fcn32 nn



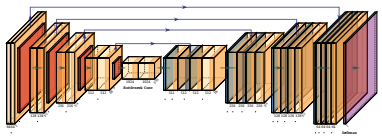
FCN-32s

nn-hed nn



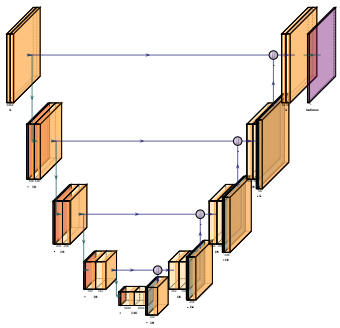
HED

nn-unet nn



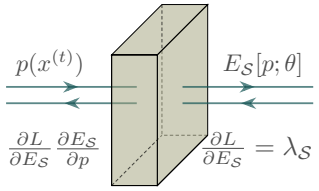
U-Net

nn-unet-ushape nn



U-Net (U shape)

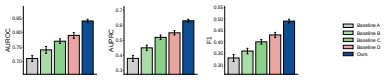
nn-softmax-loss nn



SoftmaxLoss: E_S

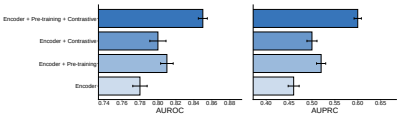
Softmax loss

f4p-bars-comparison plot



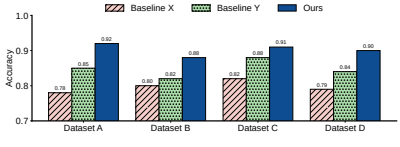
Comparison bars, one panel per metric

f4p-bars-ablation plot



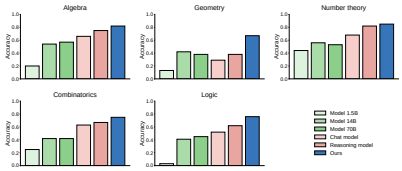
Ablation bars (opacity ramp)

f4p-bars-grouped plot



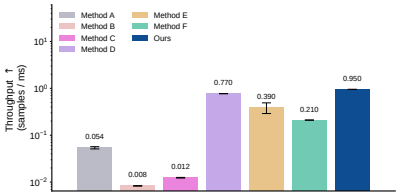
Grouped bars with hatching

f4p-bars-panels plot



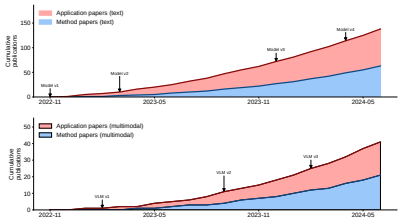
Bar panels per category

f4p-bars-log plot



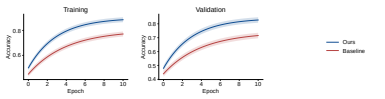
Throughput bars, log scale

f4p-trend-cumulative plot



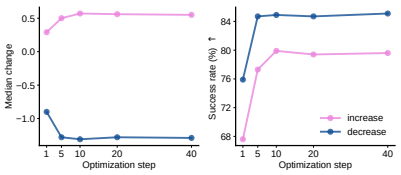
Cumulative trend with events

f4p-trend-panels plot



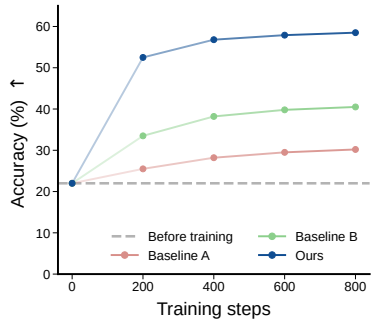
Train/validation curves + legend panel

f4p-lines-sweep plot



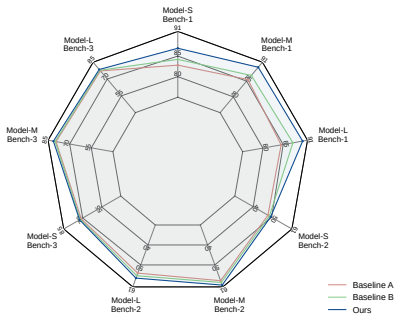
Hyper-parameter sweep lines

f4p-lines-gradient plot



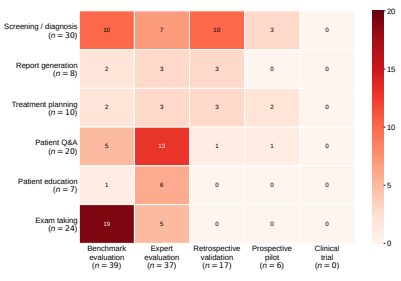
Fading training curves

f4p-radar plot



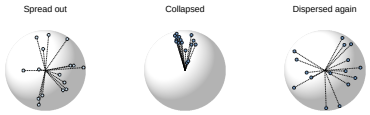
Radar chart, per-benchmark scales

f4p-heatmap plot



Annotated count heatmap

f4p-sphere plot



Points on a shaded sphere (concept)